short run aggregate supply (SRAS) curve would be upward-sloping. The long run aggregate supply curve (LRAS), however, will be vertical. Any change in the production capacity of the economy will shift the LRAS curve.

In Fig. 3.9 we depict the AD curve and an upward-sloping AS curve. The equilibrium in the economy is described by point E corresponding to output Y_0 and prices P_0. Suppose there is a favourbale demand shock to the economy so that the AD curve shifts to AD_1. As a result, there are increases in both output and prices. The new equilibrium described by point F corresponds to output Y_1 and prices P_1. The increase in output however will be temporary and output will be back at the LRAS level in the long run.

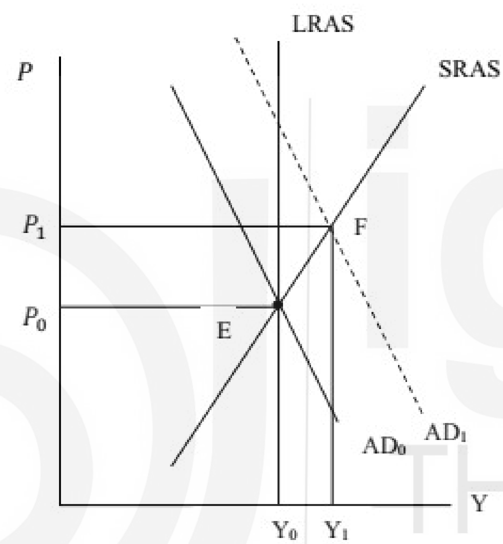

Fig. 3.9: Impact of Demand Shock

Check Your Progress 3

1) Explain how the AD curve is derived?

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2) Point out the factors that influence the position and slope of the AD curve.

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3.6 LET US SUM UP

In this Unit we described the IS-LM model, which signify the equilibrium in the real sector and the monetary sector of the economy. We derived the IS curve – on each and every point of the IS curve the real sector is in equilibrium. On the left side of the IS curve there is excess demand for goods while on the right side there is excess supply. The LM curve is the combination of all such points where the monetary sector of the economy is in equilibrium. On the right side of the LM curve, there is excess demand for money while on the left side there is excess supply of money. If the economy is operating at a point away from the IS and LM curves, it has a tendency to move towards the equilibrium point.

We derived the AD curve on the basis of IS-LM model. The AD curve traces the combinations of equilibrium output and prices. The equilibrium output and prices in the economy derived from the AS-AD model is also discussed.

3.7 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) It will affect both the intercept and slope of the IS curve. Look into equation (3.4). Decrease in the marginal propensity to consume (coefficient 'b') will lead to a decrease in the value of the intercept term. Look into the last term of the equation. It will increase the value of $\frac{e}{(1-b-d)}$. Thus, the IS curve will be steeper.

2) (i) Decrease in autonomous government expenditure will lead to a parallel downward shift in the IS curve to the left.

   (ii) Decrease in lumpsum tax will lead to a parallel upward shift in the IS curve. But the question is about decrease in tax rate, which depends on income (Y). Thus, it will affect the slope of the IS curve. The IS curve will become flatter (why? You can take a tax function $T = t_0 + t_1 Y$, substitute it in place of T in equation (3.4) and find out).

Check Your Progress 2

1) The LM curve is derived from the equality between the demand for and supply of real balances. See Section 3.3, particularly Fig. 3.3.

2) When the LM curve is vertical, demand for money is insensitive to interest rate. This is a case of liquidity trap situation.

3) Go through Sub-Section 3.4.3 and answer.

Check Your Progress 3

1) The AD curve is derived from the IS-LM model. Go through Sub-Section 3.5.1 and explain Fig. 3.8.

2) Since the AD curve is derived from the IS-LM model, it depends on the factors that influence the IS and LM curves. Explain the impact of these factors on the AD curve.

UNIT 4 OPEN ECONOMY MACROECONOMICS -I*

Structure

4.0 OBJECTIVES

After going through this Unit you should be in a position to

- explain the national income identities in an open economy;
- assess the relation between aggregate expenditure and income in an open economy;
- explain the relation between private savings-investment gap, fiscal deficit and current account deficit;
- explain the concept of Balance of Payments (BOP);
- explain current and Capital accounts of BOP;
- explain the concepts of nominal and real exchange rates; and
- explain fixed and flexible exchange rate regimes.

4.1 INTRODUCTION

In this Unit we will examine the national income identities, with a view to identifying certain fundamental differences between open and closed economies. Thereafter we examine certain basic concepts relating to Balance of Payments (BOP) and exchange rates. The BOP accounts record the transactions of an economy with the 'rest of the world' and provide important insights into the pattern of international trade and capital flows. The exchange rate is also an important variable in an open economy with important implications for international trade and competitiveness.

4.1 NATIONAL INCOME IDENTITY IN AN OPEN ECONOMY

In an open economy domestic agents have manifold economic interactions with foreigners; e.g., goods are sold to, as well as purchased from foreigners. The sale of goods and services to foreigners are exports (X) of the country and purchases from foreigners constitute imports (M). In an open economy, aggregate expenditure on final goods by domestic agents (households, firms and

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government) given by [C + I + G], includes expenditure on domestic as well as imported goods. Aggregate spending on *domestic* goods by domestic agents is given by [C + I + G – M]. To this we add expenditure on domestic goods by foreigners or exports, to obtain *total* final expenditure on domestically produced goods, i.e., [C + I + G + X –M].

From the equivalence between the expenditure, income and product approaches to measurement of aggregate output, we can say that aggregate final expenditure is equal to aggregate output produced in the economy or GDP (Y).

So the national income identity for an open economy can be written as:

$Y = C + I + G + X - M$

or, $Y = C + I + G + NX$ (where $NX = X - M$) (4.1)

Note that (X – M) is also referred to as net exports (NX) or balance of trade. When exports exceed imports, NX is positive and there is a trade surplus; when imports exceed exports, NX is negative and there is a trade deficit.

In an open economy we draw a distinction between GDP and GNP (Gross National Product). While GDP refers to aggregate incomes earned (by domestic residents and foreigners) within the geographical boundaries of the country, GNP is defined as the aggregate income of citizens of the country, irrespective of whether they are located within the country or abroad. E.g., the salary of a German consultant located in New Delhi would be a part of German GNP and India's GDP; the salary of an Indian worker in Singapore would be a part of Singapore's GDP and India's GNP.

So we define net factor income from abroad (NFIA) as follows:

NFIA = Income earned abroad by domestic factors of production *minus* income earned by foreign factors of production in the domestic economy.

To obtain GNP, NFIA is added to GDP (denoted often as Y in macroeconomics):

i.e., $GNP = Y + NFIA$ (4.2)

Note that NFIA can be positive or negative and the difference between GDP and GNP can often bequite large.For example, GDP may exceed GNP ($NFIA < 0$) in countries where mostproduction units are owned by foreign multinationals (so that a large share of the incomes generated within the economy, accrue to foreigners) or which have a sizeable migrant population employed abroad.

Using the national income identity we can show in an open economy aggregate expenditure (C+I+G) can exceedaggregate income (GNP) and that this imbalance is reflected in a current account deficit.

We can also write (4.2) as:

$GNP = C+I+G+NX+NFIA$

or, $GNP = C+I+G+CA$, (where $CA = NX + NFIA$)

or, $GNP - (C+I+G) = CA$ (4.3)

Note that CA represents the 'current account balance' of a country, which may be positive (current account surplus) or negative (current account deficit).

The identity (4.3) indicates that aggregate domestic absorption (C+I+G) exceeds aggregate income (GNP) in a country that has a current account deficit (CA < 0). Such a country is a net debtor (or, borrower) as it has to borrow to bridge the gap between income and expenditure whereas aggregate income exceeds expenditure in a country running a current account surplus (CA > 0) and it is a net lender.

A current account deficit may be financed by borrowing from the rest of the world or by sale of foreign assets held by domestic agents. We will have more to say on this in the next section on balance of payments.

The national income identity can be further used to understand whether the real sector imbalance reflected in a current account deficit, stems mainly from the private sector, or the government sector, or both.

From the income side, we know that aggregate income is either consumed (C), or saved (S) or paid out in taxes (T), so that,

$$C + S + T = GNP \tag{4.4}$$

Combining (4.3) and (4.4), we can write,

$$C + S + T = GNP = C+I+G+CA$$

or, $C + S + T = C+I+G+CA$,

or, $S + T = I+G+CA$

$$\text{or, } (S - I) + (T - G) = CA \tag{4.5}$$

From identity (4.5) it is clear that a current account deficit (CA <0) reflects either an excess of private investments over private savings (i.e., $(S - I) < 0$) or, a budget deficit (i.e., $(T–G) < 0$) or, both. Also, if $(S - I) > 0$ but, $(T–G) < 0$ and absolute value of (T–G) is larger than that of (S – I), there would be a current account deficit. The identity (4.5) represents the 'twin deficits', where an external deficit (CA < 0) is reflected as a deficit in the private sector (i.e., $(S - I) < 0$), or in the government sector (i.e., $(T–G) < 0$), or in both.

The national income identities clearly indicate that an imbalance in the current account reflects an imbalance in the real economy.That is, whenever a country runs a current account deficit, there is an excess of aggregate expenditure over income; this may occur because, private savings are too low, or there is aprivate investment boom, or, the government is running a budget deficit or some combination of these factors. Conversely, aggregate income exceeds expenditure in a country that runs a current account surplus and either its private savings would exceed investments (i.e. $(S - I) > 0$) or, public savings would be positive (i.e., $(T - G) > 0$) or, both.

The main difference between closed and open economies should be evident from the above discussion. In a closed economy, by definition, there are no transactions with the 'rest of the world'and CA is zero. Thus aggregate expenditure cannot exceed aggregate income. It means, an excess of private investments over savings ($S - I < 0$) needs to be compensated by public savings ($T–G > 0$). However, open economies can borrow from the rest of the world. Thus it can sustain excess of expenditure over income and excess of investments (I) over private (S) and public savings ($T - G$). International trade and capital flows allow countries running current account deficits to draw on the savings of other countries running current account surpluses.

Check Your Progress 1

1) Use the national income identity for an open economy and show that in a country running a trade deficit (i.e., $NX < 0$): (a) aggregate expenditure must be greater than GDP, and (b) national savings (i.e., sum of private savings and public savings) must be less than investments.

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2) Use the national income identity to bring out at least two fundamental differences between open and closed economies.

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4.3 BALANCE OF PAYMENTS AND EXCHANGE RATE

The Balance of Payments (BOP) of a country is a record of all its transactions with foreigners during a given period (typically one year). The transactions include those involving purchase and sale of goods, services, physical and financial assets, as well as transfer payments.

4.3.1 Balance of Payments

The BOP account has two main components, viz., the current account and the capital account.

Current Account: The current account of BOP records receipts from and payments to foreigners due to international trade in goods and services (including factor services). In the current account transactions involving *inflows* of foreign exchange (e.g. exports) are a credit and those involving *outflows* of foreign exchange (e.g., imports) are a debit entry. The current account comprises two main components: the balance of trade and the balance on invisibles. Transactions involving trade in services, factor incomes and transfers are also referred to as 'invisibles' in the current account. These are further explained below.

The balance of trade or merchandise trade balance is the difference between exports and imports of goods by a country and it may be positive, negative or zero. A negative balance of trade or a trade deficit indicates that the country's imports of goods exceed exports; a positive trade balance indicates the country has a trade surplus, with exports of goods exceeding imports.

Trade in services (e.g., India's software exports) is a part of invisibles in the current account. The balance of trade in services captures the difference between a country's service exports and imports and it may also be positive or negative, depending on whether the country is a net exporter or net importer of services.

Factor incomes include *net investment incomefrom abroad* which measures investment earnings from abroad earned by Indians *minus* foreigners' earnings from their Indian assets. Indian residents may receive investment income, such as dividends and interest, on the foreign assets they hold while foreign residents receive investment income on the Indian assets they possess. The former is a credit (inflow of foreign exchange) while the latter is a debit entry (outflow of foreign exchange) in the current account.

Unilateral Transfers, that are overseas payments made without any *quid pro quo*, are also recorded in the current account. These include assistance by foreign governments during war or natural calamity (foreign aid), workers' remittances (e.g., money sent by overseas Indians to their families in India), personal gifts, donations etc. The *net unilateral transfers* are transfers received by Indians from abroad minus similar transfers sent to foreigners from India. For many developing countries workers' remittances are an important source of foreign exchange earnings.

The current account balance is the sum of (i) the balance on merchandise trade and (ii) the balance on invisibles (which comprises balance on services trade, net investment income and net transfers). When total foreign exchange receipts on account of all the transactions recorded in the current account are greater than total payments, the country has a current account surplus. In case the foreign exchange payments exceed the receipts, the country has a current account deficit.

Note that even if a country has a trade deficit it could have a current account surplus, if it has a positive and very high net invisibles balance. This can be seen in many countries that are net recipients of large unilateral transfers and remittances. For example, remittances received from Indian workers employed abroad are an important source of foreign exchange earnings for India. India is the largest recipient of remittances in absolute figure ($87 billion in 2021) while in terms of percentage of GDP Lebanon is the largest recipient that contributed 54 per cent of GDP in 2021.

Capital Account: The capital account of the BOP includes transactions involving cross-border purchase and sale of real and financial assets. The asset in question could be physical assets like land, factories, houses or financial assets like stocks and bonds. In the capital account each transaction is recorded twice, once as a credit and once as a debit entry, thereby capturing the two aspects of each transaction. E.g., purchase of a foreign financial asset such as foreign government bond or shares of a foreign firm by a domestic resident involves an asset import and a capital outflow (payment made by the domestic resident for the foreign asset).

Broadly two types of capital flows are recorded in the capital account – debt creating and non-debt creating flows. For example, when domestic agents borrow from foreign financial institutions, the country experiences a debt creating capital inflow, as this loan has to be repaid at a future date. However, foreign firms investing in the domestic country (e.g., foreign direct investment or FDI), involve a non-debt creating capital inflow, as this does not create any commitment for repayment for the recipient nation. Capital flows can also be classified on the basis of the duration involved. For example, foreign institutional investors (FIIs) purchasing shares of domestic firms, involves a short term capital inflow (also known as foreign portfolio investment or FPI). The FIIs may sell the equity and withdraw their funds at a short notice. Such withdrawal of foreign funds would involve a capital outflow. Typically, investment by foreign firms in country's production sector (e.g., Greenfield FDI) involves longer term capital inflows that are not likely to be withdrawn at short notice unlike investment in the financial sector (e.g., FPI).

When capital inflows exceed capital outflows, the country has a capital account surplus. In the opposite case, when outflows exceed inflows it has a capital account deficit.

In the current account, in general, transactions involving the inflow of foreign exchange are recorded as a credit entry, with the corresponding debit entry appearing in the capital account. Therefore a surplus (or deficit) in current account is mirrored in a corresponding deficit (or surplus) in the capital account. If a country has a current account deficit, this must be financed either by selling assets or by borrowing, so there must be a corresponding surplus in the capital account. So a current account deficit is financed by net capital inflows, either on account of the private sector or the government sector or both.

Overall BOP Balance: Every transaction is recorded twice in the BOP account, once as a debit and once as credit, as per the principles of double entry bookkeeping. Therefore in an accounting sense the BOP is always balanced. However, in practice, this is not observed, primarily because of problems related to data collection. There are time lags involved in the international transactions; e.g., data on exports is collected from customs, at the time goods are shipped, whereas payments for the same may be received six months later and hence may not be recorded in BOP of that year. Therefore, in order to balance the BOP account an entry for *errors and omissions* is included.

Even though the BOP always balances, we can talk about an overall BOP surplus or deficit. Overall BOP balance is obtained by adding the current and capital account balances. A country has a BOP surplus in the following cases: (i) there is current account surplus as well as capital account surplus; (ii) there is a current account deficit but a capital account surplus that is larger in magnitude; (iii) there is a current account surplus and a relatively smaller capital account deficit. There would be a BOP deficit, either when there is a deficit on both current and capital accounts or when there is a relatively large deficit in either one of the current or capital accounts.

In the case of a BOP surplus there is an increase in the foreign exchange reserves of the country. In the case of a deficit, the stock of foreign exchange reserves is depleted. Let us discuss this issue in details.

Change in Foreign Exchange Reserves

Apart from transactions made by private and government agents, the capital account of the BOP also includes *official reserve transactions* that involve change in the stock of foreign exchange reserves held by the central bank. When there is a BOP surplus, the country experiences a net inflow of foreign exchange, leading to a correspondingincrease in foreign exchange reserves. In the case of a BOP deficit, there is net outflow of foreign exchange and a corresponding reduction in foreign exchange reserves. Thus,

BOP Balance = Current Account + Capital Account (including errors and omissions) + Official Reserve Transactions = 0

In case a country has a current account deficit (of say, – $15,000), but a capital account surplus that is smaller in magnitude (say, $12,000), then it has a BOP deficit (of –$3000) that leads to a decrease in foreign exchange reserves (by $3000).

Note that a BOP surplus is recorded with a positive sign in the BOP, while the corresponding *increase* in reserves has a *negative* sign, so that the two add up to zero. Therefore in BOP accounts an increase in foreign exchange reserves appears with a negative sign whereas a decrease in reserves has a positive sign.

4.3.2 Exchange Rate

The exchange rate is an important concept in an open economysuch as India, that has transactions involving foreign goods, services and assets, whose prices are denominated in terms of foreign currencies. Throughout the following discussion we assume that the domestic country is India and the foreign country is the USA.

Demand for foreign exchange arises whenever domestic agents purchase foreign goods, services or assets (i.e., imports and capital outflows); whereas, sale of domestic goods, services and assets to foreigners (i.e., exports and capital